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Inventor(s)	Tu; Wen-Hsien et al.

Integrated circuit device

Abstract

An integrated circuit device is provided. The integrated circuit device includes a semiconductor substrate, first and second semiconductor fins over the semiconductor substrate, and first and second epitaxy structures respectively on the first and second semiconductor fins. The first epitaxy structure is merged with the second epitaxy structure, and a bottom surface of the second epitaxy structure is lower than a bottom surface of the first epitaxy structure.

Inventors: **Tu; Wen-Hsien (New Taipei, TW), Ke; Dong-Jie (Taichung, TW)**

Applicant: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.** (Hsinchu, TW)

Family ID: **1000008818588**

Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD. (Hsinchu, TW)**

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Primary Examiner: Eskridge; Cory W

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a divisional application of U.S. patent application Ser. No. 17/460,659, filed Aug. 30, 2021, the entirety of which is incorporated by reference herein in its entirety.

BACKGROUND
(1) As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance, and lower costs, challenges from both fabrication and design issues have resulted in the development of three dimensional designs, such as a fin-like field effect transistor (FinFET). A FinFET includes an extended semiconductor fin that is elevated above a substrate in a direction normal to the plane of the substrate. The channel of the FET is formed in this vertical fin. A gate is provided over (e.g., wrapping) the fin. The FinFETs further can reduce the short channel effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1** is a cross-sectional view of an integrated circuit device according to some embodiments of the present disclosure.

(3) FIG. **2** is a cross-sectional view of an integrated circuit device according to some embodiments of the present disclosure.

(4) FIG. **3** is a cross-sectional view of an integrated circuit device according to some embodiments of the present disclosure.

(5) FIG. **4** is a cross-sectional view of an integrated circuit device according to some embodiments of the present disclosure.

(6) FIG. **5** is a cross-sectional view of an integrated circuit device according to some embodiments of the present disclosure.

(7) FIGS. **6**A to **14**D illustrate a method for manufacturing an integrated circuit device at various stages in accordance with some embodiments of the present disclosure.

(8) FIGS. **15**A to **20**D illustrate a method for manufacturing an integrated circuit device at various stages in accordance with some embodiments of the present disclosure.

(9) FIG. **21** is a cross-sectional view of an integrated circuit device in accordance with some embodiments of the present disclosure.

(10) FIGS. **22** to **27**D illustrate a method for manufacturing an integrated circuit device at various stages in accordance with some embodiments of the present disclosure.

(11) FIG. **28** is a cross-sectional view of an integrated circuit device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) The present disclosure is directed to, but not otherwise limited to, an integrated circuit device having a FinFET device. The FinFET device, for example, may be a complementary metal-oxide-semiconductor (CMOS) device comprising a P-type metal-oxide-semiconductor (PMOS) FinFET device and an N-type metal-oxide-semiconductor (NMOS) FinFET device. The following disclosure will continue with a FinFET example to illustrate various embodiments of the present

disclosure. It is understood, however, that the application should not be limited to a particular type of device, except as specifically claimed.

(15) The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

(16) FIG. **1** is a cross-sectional view of an integrated circuit device **100** according to some embodiments of the present disclosure. The integrated circuit device **100** includes a semiconductor substrate **110**, isolation dielectrics **120**, a gate structure DG, gate spacers **150**, a fin sidewall spacer **162**, and an epitaxial feature **170**. The isolation dielectrics **120** is over the substrate **110** and surrounds a semiconductor fin **112** of the semiconductor substrate **110**. The gate structure DG is crossing a first portion of the fin **112** protruding from a top surface of the isolation dielectrics **120**. The gate structure DG may include a gate dielectric layer **142** and a gate electrode layer **144**. The gate spacers **150** are respectively on opposite sides of the gate structure DG. A second portion of the fin **112** not covered by the gate structure DG may be recessed to form a recess **112***r*. The epitaxial feature **170** may be formed over the recessed second portion of the fin **112** and filling the recess **112***r*. In some embodiments of the present disclosure, the fin sidewall spacer **162** is at a first side **170***a* of the epitaxial feature **170**, and a second side **170***b* of the epitaxial feature **170** may be free of a fin sidewall spacer. Through the configuration of the fin sidewall spacer **162**, the epitaxial growth of the epitaxial feature **170** on the second portion of the fin **112** may be limited by the fin sidewall spacer **162**. For example, the epitaxial feature **170** extends further to the second side **170***b* than to the first side **170***a*.

(17) FIG. **2** is a cross-sectional view of an integrated circuit device **100** according to some embodiments of the present disclosure. The present embodiments are similar to those shown in FIG. **1**, and one of the differences between the present embodiments and the embodiments of FIG. **1** is that: epitaxial features **172** and **174** are respectively on semiconductor fins **112***a* and **112***b* of the semiconductor substrate **110**, and the fin sidewall spacer **162** is at a first side **172***a* of the epitaxial feature **172**. In some embodiments of the present disclosure, a second side **172***b* of the epitaxial feature **172** and two opposite sides **174***a* and **174***b* of the epitaxial feature **174** are free of a fin sidewall spacer. Through the configuration of the fin sidewall spacer **162**, the epitaxial feature **172** extends further to the epitaxial feature **174**, thereby merging with the epitaxial feature **174**. Other details are similar to those illustrated with the embodiments of FIG. **1**, and therefore not repeated herein.

(18) FIG. **3** is a cross-sectional view of an integrated circuit device **100** according to some embodiments of the present disclosure. The present embodiments are similar to those shown in FIG. **2**, and one of the differences between the present embodiments and the embodiments of FIG. **2** is that: epitaxial features **171-178** are respectively on semiconductor fins **112***a*-**112***h* of the semiconductor substrate **110**, and the fin sidewall spacers **162** are respectively at a first side **171***a* of the epitaxial feature **171**, a second side **174***b* of the epitaxial feature **174**, a first side **175***a* of the epitaxial feature **175**, and a second side **178***b* of the epitaxial feature **178**. Through the configuration of the fin sidewall spacers **162**, the epitaxial features **171-174** are merged, the epitaxial features **175-178** are merged, and the merged epitaxial features **171-174** are spaced apart from the merged epitaxial features **175-178**.

(19) In some embodiments of the present disclosure, the integrated circuit device **100** further includes fin sidewall spacers **162**′ on opposite sides of the epitaxial features **172**, **173**, **176**, and

177, and on a second side **171***b* of the epitaxial feature **171**, a first side **174***a* of the epitaxial feature **174**, a second side **175***b* of the epitaxial feature **175**, and a first side **178***a* of the epitaxial feature **178**. In some embodiments, a height of the fin sidewall spacers **162′** is less than a height of the fin sidewall spacers **162**, such that the fin sidewall spacers **162′** have less influence on the epitaxial growth of the epitaxial features **171-178**, which in turn result in that the epitaxial features **171-174** are merged, and the epitaxial features **175-178** are merged. In some other embodiments, the fin sidewall spacers **162′** may be omitted.

(20) FIG. **4** is a cross-sectional view of an integrated circuit device **100** according to some embodiments of the present disclosure. The present embodiments are similar to those shown in FIG. **2**, and one of the differences between the present embodiments and the embodiments of FIG. **2** is that: a recess **112***rb* over the semiconductor fin **112***b* is deeper than a recess **112***ra* over the semiconductor fin **112***a*. By designing the recesses **112***ra* and **112***rb* have different depths, the semiconductor material may be epitaxially deposited over the semiconductor fins **112***a* and **112***b* with different facets. For example, the epitaxial feature **174** has a sidewall surface **174***s* with (**111**) facet, which tilts substantially 45 degrees with respect to a top surface of the substrate **110**. The epitaxial feature **172** over the semiconductor fin **112***a* may have a sidewall surface **172***s* at an angle greater than 45 degrees with respect to the top surface of the substrate **110**. That is, the sidewall surface **172***s* of the epitaxial feature **172** may be more vertical than a sidewall surface **174***s* of the epitaxial feature **174** is. Through the configuration, the epitaxial feature **172** extends laterally less than the epitaxial feature **174** does, and the epitaxial feature **174** over the semiconductor fin **112***b* has a greater size than the epitaxial feature **172** over the semiconductor fin **112***a*. In some embodiments, a height of the epitaxial feature **174** may be greater than a height of the epitaxial feature **172**. In some embodiments, a top surface of the epitaxial feature **174** may be higher than a top surface of the epitaxial feature **172**. The depths of the recesses **112***ar* and **112***br* are tuned such that the epitaxial features **172** and **174** are merged. In the present embodiments, the fin sidewall spacers **162** and **162′** (referring to FIGS. **1-3**) may be omitted. Alternatively, in some embodiments, fin sidewall spacers **162** and/or **162′** (referring to FIGS. **1-3**) may be formed on sidewalls of the epitaxial features **172** and **174**.

(21) FIG. **5** is a cross-sectional view of an integrated circuit device **100** according to some embodiments of the present disclosure. The present embodiments are similar to those shown in FIG. **4**, and one of the differences between the present embodiments and the embodiments of FIG. **4** is that: epitaxial features **171-179** are respectively on semiconductor fins **112***a***-112***i* of the semiconductor substrate **110**, and recesses **112***r*, over the semiconductor fins **112***b***-112***d*, **112***g*, **112***h* are deeper than the recesses **112***r* over the semiconductor fins **112***a*, **112***e*, **112***f*, and **112***i*. Through the configuration, the epitaxial features **172-174**, **177**, and **178** have a greater size than the epitaxial features **171**, **175**, **176**, and **179**. In some embodiments, a height of the epitaxial features **172-174**, **177**, and **178** may be greater than a height of the epitaxial features **171**, **175**, **176**, and **179**. In some embodiments, a top surface of the epitaxial features **172-174**, **177**, and **178** may be higher than a top surface of the epitaxial features **171**, **175**, **176**, and **179**. The deepness of the recesses **112***r* are tuned such that the epitaxial features **171-175** are merged, the epitaxial features **176-179** are merged, and the merged epitaxial features **171-175** are spaced apart from the merged epitaxial features **176-179**. Also, in the present embodiments, the fin sidewall spacers **162** and **162′** (referring to FIGS. **1-3**) may be omitted. Alternatively, in some embodiments, fin sidewall spacers **162** and/or **162′** (referring to FIGS. **1-3**) may be formed on sidewalls of the epitaxial features **171-179**.

(22) FIGS. **6**A to **14**D illustrate a method for manufacturing an integrated circuit device **100** at various stages in accordance with some embodiments of the present disclosure. The illustration is merely exemplary and is not intended to be limiting beyond what is specifically recited in the claims that follow. It is understood that additional operations may be provided before, during, and after the operations shown by FIGS. **6**A to **14**D, and some of the operations described below can be

replaced or eliminated for additional embodiments of the method. The order of the operations/processes may be interchangeable.

(23) Reference is made to FIGS. **6**A and **6**B. FIG. **6**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **6**B is a cross-sectional view taken along line B-B of FIG. **6**A. A substrate **110** is illustrated. In some embodiments, the substrate **110** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like. The substrate **110** may be a wafer, such as a silicon wafer. Generally, an SOI substrate comprises a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, a silicon or glass substrate. For example, the substrate **110** may include a wafer **110***a*, a semiconductor device layer **110***b* over the wafer **110***a*, and a BOX layer (not shown) between the wafer **110***a* and the semiconductor device layer **110***b*. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **110** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

(24) A patterned mask P**1** is formed over the substrate **110**. The patterned mask P**1** may be a hard mask for protecting the underlying substrate **110** against subsequent etching process. The patterned mask P**1** may be formed by a series of operations including deposition, photolithography patterning, and etching processes. The photolithography patterning processes may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), and/or other applicable processes.

(25) Reference is made to FIGS. **7**A and **7**B. FIG. **7**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **7**B is a cross-sectional view taken along line B-B of FIG. **7**A. In some embodiments, the substrate **110** is etched through the patterned mask P**1** (referring to FIGS. **6**A and **6**B) to form the trenches T, and portions of the substrate **110** surrounded by the trenches T can be referred to as semiconductor fins **112**. The etching processes may include dry etching, wet etching, and/or other etching methods (e.g., reactive ion etching). In some embodiments, plural semiconductor fins **112** are formed substantially parallel to each other. For better illustration, the semiconductor fins **112** are respectively labelled as semiconductor fins **112***a*-**112***b* in the figure.

(26) Reference is made to FIGS. **8**A-**8**C. FIG. **8**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **8**B is a cross-sectional view taken along line B-B of FIG. **8**A. FIG. **8**C is a cross-sectional view taken along line C-C of FIG. **8**A. Shallow trench isolation (STI) features **130** are formed interposing the fins **112**. In the present embodiments, a dielectric layer is deposited over the substrate **110** and filling the trenches T. In some embodiments, the dielectric layer may include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric, combinations thereof, and/or other suitable materials. In various examples, the dielectric layer may be deposited by a CVD process, a sub-atmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a physical vapor deposition (PVD) process, and/or other suitable process. In some embodiments, the dielectric layer may include a multi-layer structure, for example, having one or more liner layers. After deposition of the dielectric layer, the deposited dielectric material is thinned and planarized, for example by a chemical mechanical polishing (CMP) process, thereby forming the STI features **130** between the fins **112**. The STI features **130** may be recessed by suitable etching process, such as a dry etching process, a wet etching process,

and/or a combination thereof. The recessing process provides the fins **112** extending above the STI features **130**.
(27) Reference is made to FIGS. **9**A-**9**D. FIG. **9**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **9**B is a cross-sectional view taken along line B-B of FIG. **9**A. FIG. **9**C is a cross-sectional view taken along line C-C of FIG. **9**A. FIG. **9**D is a cross-sectional view taken along line D-D of FIG. **9**A. A gate structure DG is formed over a portion of the semiconductor fin **112**. In some embodiments, the gate structure DG is dummy (sacrificial) gate structures that is subsequently removed. Thus, in some embodiments using a gate-last process, the gate structure DG is dummy gate structure and will be replaced by a final gate structure at a subsequent processing stage. In particular, the dummy gate structure DG may be replaced at a later processing stage by a high-k dielectric layer (HK) and metal gate electrode (MG) as discussed below. In some embodiments, the dummy gate structure DG is formed over the substrate **110** and is at least partially disposed over the fins **112**. The portion of the fins **112** underlying the dummy gate structure DG may be referred to as the channel region. The dummy gate structure DG may also define a source/drain (S/D) region of the fins **112**, for example, the regions of the fin **112** adjacent and on opposing sides of the channel region.
(28) In some embodiments, the dummy gate structure DG is formed by various process steps such as layer deposition, patterning, etching, as well as other suitable processing steps. In the illustrated embodiments, the formation of the gate structure DG first includes depositing a dummy gate dielectric layer **142**, a dummy gate electrode layer **144** and a hard mask layer **146** over the fins **112**, and then patterning the layers **142-146** to form the dummy gate structure DG. In some embodiments, the dummy gate dielectric layer **142** may include SiO.sub.2, silicon nitride, a high-k dielectric material and/or other suitable material. In various examples, the dummy gate dielectric layer **152** may be deposited by a CVD process, a sub-atmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a PVD process, or other suitable process. By way of example, the dummy gate dielectric layer **142** may be used to prevent damages to the fins **112** by subsequent processes (e.g., subsequent formation of the dummy gate structure). In some embodiments, the dummy gate electrode layer **144** may include polycrystalline silicon (polysilicon). In some embodiments, the hard mask layer **146** includes an oxide layer such as a pad oxide layer that may include SiO.sub.2, and a nitride layer such as a pad nitride layer that may include Si.sub.3N.sub.4 and/or silicon oxynitride. Exemplary layer deposition processes include CVD (including both low-pressure CVD and plasma-enhanced CVD), PVD, ALD, thermal oxidation, e-beam evaporation, or other suitable deposition techniques, or combinations thereof.
(29) In forming the gate structure for example, the patterning process includes a lithography process (e.g., photolithography or e-beam lithography) which may further include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, photoresist developing, rinsing, drying (e.g., spin-drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. In some embodiments, the etching process may include dry etching (e.g., RIE etching), wet etching, and/or other etching methods. In some embodiments, after patterning the dummy gate electrode layer **144**, the dummy gate dielectric layer **142** is removed from the S/D regions of the fins **112**. The etch process may include a wet etch, a dry etch, and/or a combination thereof. The etch process is chosen to selectively etch the dummy gate dielectric layer **142** without substantially etching the fins **112**, the dummy gate electrode layer **144**, and the hard mask **146**.
(30) In some embodiments, gate spacers **150** are then formed on opposite sidewalls of the dummy gate structure DG. In some embodiments, the gate spacers **150** may include silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, silicon oxycarbide, porous dielectric materials, hydrogen doped silicon oxycarbide (SiOC:H), low-k dielectric materials or other suitable dielectric material. The gate spacers **150** may include a single

layer or multilayer structure made of different dielectric materials. The method of forming the gate spacers **150** includes blanket forming a dielectric layer on top surface and sidewalls of the dummy gate structure DG using, for example, CVD, PVD or ALD, and then performing an etching process such as anisotropic etching to remove horizontal portions of the dielectric layer. The remaining portions of the dielectric layer on sidewalls of the dummy gate structure DG can serve as the gate spacers **150**. In some embodiments, the gate spacers **150** may be used to offset subsequently formed doped regions, such as source/drain regions. The gate spacers **150** may further be used for designing or modifying the source/drain region profile.
(31) Reference is made to FIG. **10**. FIG. **10** is a cross-sectional view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure, and FIG. **10** is taken along the same cut as in FIG. **9**B. A spacer material layer **160** is conformally deposited over the structure of FIGS. **9**A-**9**D. For example, the spacer material layer **160** extends over top surfaces of the STI features **130**, sidewalls of the fins **112***a* and **112***b*, and top surfaces of the fins **112***a* and **112***b*. The spacer material layer **160** may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN films, silicon oxycarbide, SiOCN films, and/or combinations thereof. In some embodiments, the spacer material layer **160** may include multiple layers, such as main spacer walls, liner layers, and the like. By way of example, the spacer material layer **160** may be formed by depositing a dielectric material using processes such as, CVD process, a sub-atmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a PVD process, or other suitable process.
(32) In some embodiments, a patterned mask P**2** is formed over a portion of the spacer material layer **160**. The patterned mask P**2** may be a photoresist for protecting the spacer material layer **160** against subsequent etching process. The patterned mask P**2** may be formed by photolithography patterning processes, including photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), and/or other applicable processes. In some other embodiments, the patterned mask P**2** may be a hard mask for protecting the spacer material layer **160** against subsequent etching process. The hard mask may include Si.sub.3N.sub.4 and/or silicon oxynitride.
(33) Reference is made to FIGS. **11**A-**11**C. FIG. **11**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **11**B is a cross-sectional view taken along line B-B of FIG. **11**A. FIG. **11**C is a cross-sectional view taken along line C-C of FIG. **11**A. The spacer material layer **160** (referring to FIG. **10**) is patterned to form a spacer **161** on a sidewall of the fin **112***a* facing the fin **112***b*, and a spacer **162** on a sidewall of the fin **112***b* facing the fin **112***a*, thereby exposing portions of the semiconductor fins **112***a* and **112***b*. In other words, the spacers **161-162** are between the semiconductor fins **112***a* and **112***b*. The patterning includes a suitable etching process. In some embodiments, the etching process includes a dry etching process using an etchant including a fluorine-containing gas, a chlorine-containing gas, other etching gas, or a combination thereof, such as CF.sub.4, SF.sub.6, NF.sub.3, or Cl.sub.2. By the etching process, portions of the spacer material layer **160** (referring to FIG. **10**) exposed by the patterned mask P**2** (referring to FIG. **10**) is etched away, while a portion of the spacer material layer **160** covered by the patterned mask P**2** (referring to FIG. **10**) is protected from being etched. The remaining portion of the spacer material layer **160** (referring to FIG. **10**) forms the spacers **161-162**. In the present embodiments, the remaining portion of the spacer material layer **160** (referring to FIG. **10**) further forms a portion **169** extending horizontally over the STI features **130** and connecting between the spacers **161** and **162**. A combination of the spacers **161**, **162**, and the portion **169** may be referred to as spacer **160**′. In some other embodiments, the patterned mask P**2** (referring to FIG. **10**) may be designed in another way such that the portion **169** is etched away, and the spacer **161** may be spaced apart from the spacer **162**. After the patterning process, the patterned mask P**2** (referring to FIG. **10**) may be removed by suitable stripping process.

(34) Reference is made to FIGS. **12**A-**12**C. FIG. **12**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **12**B is a cross-sectional view taken along line B-B of FIG. **12**A. FIG. **12**C is a cross-sectional view taken along line C-C of FIG. **12**A. Exposed portions of the semiconductor fins **112** (e.g., the fins **112***a* and **112***b*) are recessed by suitable etching process by using the dummy gate structure DG, the gate spacers **150**, and the STI features **130** as an etch mask, resulting in recesses **112***r* into the semiconductor fins **112**. In some embodiments, the etching process may be a dry etching, a wet etch, or the combination thereof. In some embodiments, the etching process includes a dry etching process using an etchant including a halogen-containing compounds or the like. In some embodiments, the etching process may also consume the spacer **160**′ (referring to FIGS. **11**A-**11**C). For example, in some embodiments, the portion **169** of the spacer **160**′ (referring to FIGS. **11**A-**11**C) may be removed by the etching process, such that the spacer **161** and **162** are disconnected from each other. For example, in some embodiments, the etching process may also lower top surfaces of the spacers **161** and **162**.
(35) Reference is made to FIGS. **13**A-**13**C. FIG. **13**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **13**B is a cross-sectional view taken along line B-B of FIG. **13**A. FIG. **13**C is a cross-sectional view taken along line C-C of FIG. **13**A. Epitaxial features **171** and **172** are respectively formed over the exposed portions of the semiconductor fins **112***a* and **112***b*. In some embodiments, the epitaxial features **171** and **172** may include Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, or other suitable material. The epitaxial features **171** and **172** may be in-situ doped during the epitaxial process by introducing doping species including: p-type dopants, such as boron or BF.sub.2; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. If the epitaxial features **171** and **172** are not in-situ doped, an implantation process (i.e., a junction implant process) is performed to dope the epitaxial features **171** and **172**.
(36) The epitaxial features **171** and **172** may be formed by performing an epitaxial growth process that provides an epitaxial material on the fins **112**. Suitable epitaxial processes include CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy, and/or other suitable processes. The epitaxial growth process may use gaseous and/or liquid precursors, which interact with the composition of semiconductor materials of the fins **112**.
(37) In some further embodiments, the epitaxial growths of the epitaxial features **171** and **172** are respectively confined by the fin sidewall spacers **161** and **162**, thereby preventing the epitaxial features **171** and **172** from merging with each other. After the epitaxial growths of the epitaxial features **171** and **172**, the fin sidewall spacers **161** and **162** are respectively formed on a side of the epitaxial feature **171** and a side of the epitaxial feature **172** facing each other.
(38) Reference is made to FIGS. **14**A-**14**C. FIG. **14**A is a schematic top view of the integrated circuit device **100** according to some embodiments of the present disclosure. FIG. **14**B is a cross-sectional view taken along line B-B of FIG. **14**A. FIG. **14**C is a cross-sectional view taken along line C-C of FIG. **14**A. FIG. **14**D is a cross-sectional view taken along line C-C of FIG. **14**D. An ILD layer **180** is formed on the substrate **110**. In some embodiments, the ILD layer **180** includes materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG). The ILD layer **180** may be deposited by a PECVD process or other suitable deposition technique. In some embodiments, a contact etch stop layer (CESL) is also formed prior to forming the ILD layer **180**. In some examples, the CESL includes a silicon nitride layer, silicon oxide layer, a silicon oxynitride layer, and/or other suitable materials having a different etch selectivity than the ILD layer **180**. In some embodiments, after formation of the ILD layer **180**, the integrated circuit device **100** may be subject to a high thermal budget

process to anneal the ILD layer **180**.

(39) In some examples, after depositing the ILD layer **180**, a planarization process may be performed to remove excessive materials of the ILD layer **180**. For example, a planarization process includes a chemical mechanical planarization (CMP) process which removes portions of the ILD layer **180** (and CESL layer, if present) overlying the dummy gate structures DG and planarizes a top surface of the integrated circuit device **100**. In some embodiments, the CMP process also removes hard mask layers **136** (as shown in FIGS. **13**B-**13**C) and exposes the dummy gate electrode layer **144**.

(40) Subsequently, the dummy gate structure DG is replaced with the metal gate structure **190**. For example, the dummy gate structure DG is removed by a selective etching process (e.g., selective dry etching, selective wet etching, or a combination thereof) that etches the materials in dummy gate structures DG at a faster etch rate than it etches other materials (e.g., gate spacers **150**, CESL and/or ILD layer **180**), thus resulting in a gate trench GT between corresponding gate spacers **150**. Then, a metal gate structure **190** is formed in the gate trench GT. The metal gate structure **190** may be a high-k/metal gate stack, however other compositions are possible. In various embodiments, the metal gate structure **190** includes an interfacial layer **192**, a high-k dielectric layer **194**, a work function metal layer **196**, and a fill metal **198** filling a remainder of gate trenches GT. Formation of the high-k/metal gate structures **430** may include depositions to form various gate materials, one or more liner layers, and one or more CMP processes to remove excessive gate materials.

(41) In some embodiments, the interfacial layer **192** may include a dielectric material such as silicon oxide (SiO_2), HfSiO, or silicon oxynitride (SiON). The interfacial layer **192** may be formed by chemical oxidation, thermal oxidation, atomic layer deposition (ALD), chemical vapor deposition (CVD), and/or other suitable method. The high-k dielectric layer **194** may include hafnium oxide (HfO_2). Alternatively, the high-k dielectric layer **194** may include other high-k dielectrics, such as hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), lanthanum oxide (LaO), zirconium oxide (ZrO), titanium oxide (TiO), tantalum oxide (Ta_2O_5), yttrium oxide (Y_2O_3), strontium titanium oxide ($SrTiO_3$, STO), barium titanium oxide ($BaTiO_3$, BTO), barium zirconium oxide (BaZrO), hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), aluminum oxide (Al_2O_3), silicon nitride (Si_3N_4), oxynitrides (SiON), and combinations thereof.

(42) The work function metal layer **196** may include work function metals to provide a suitable work function for the high-k/metal gate structures **190**. For an n-type GAA FET, the work function metal layer **196** may include one or more n-type work function metals (N-metal). The n-type work function metals may exemplarily include, but are not limited to, titanium aluminide (TiAl), titanium aluminium nitride (TiAlN), carbo-nitride tantalum (TaCN), hafnium (Hf), zirconium (Zr), titanium (Ti), tantalum (Ta), aluminum (Al), metal carbides (e.g., hafnium carbide (HfC), zirconium carbide (ZrC), titanium carbide (TiC), aluminum carbide (AlC)), aluminides, and/or other suitable materials. On the other hand, for a p-type GAA FET, the work function metal layer **196** may include one or more p-type work function metals (P-metal). The p-type work function metals may exemplarily include, but are not limited to, titanium nitride (TiN), tungsten nitride (WN), tungsten (W), ruthenium (Ru), palladium (Pd), platinum (Pt), cobalt (Co), nickel (Ni), conductive metal oxides, and/or other suitable materials.

(43) In some embodiments, the fill metal **198** may exemplarily include, but are not limited to, tungsten, aluminum, copper, nickel, cobalt, titanium, tantalum, titanium nitride, tantalum nitride, nickel silicide, cobalt silicide, TaC, TaSiN, TaCN, TiAl, TiAlN, or other suitable materials.

(44) In some embodiments, contacts **210** are formed in the ILD layer **180** and over the epitaxial features **171** and **172**. In some embodiments, contact openings are first formed through the ILD layer **180** to expose the epitaxial features **171** and **172** by using suitable photolithography and

etching techniques. Subsequently, silicide regions **200** are formed on the front side of the epitaxial features **171** and **172** by using a silicidation process, followed by forming contacts **210** over the silicide regions **200**. Silicidation may be formed by depositing a metal layer (e.g., nickel layer or cobalt layer) over the exposed epitaxial features **171** and **172**, annealing the metal layer such that the metal layer reacts with silicon (and germanium if present) in the epitaxial features **171** and **172** to form the metal silicide region **200** (e.g., nickel silicide or cobalt silicide), and thereafter removing the non-reacted metal layer. The contact **210** may be formed by depositing one or more metal materials (e.g., tungsten, cobalt, copper, the like or combinations thereof) to fill the contact holes by using suitable deposition techniques (e.g., CVD, PVD, ALD, the like or combinations thereof), followed by a CMP process to remove excess metal materials outside the contact openings.

(45) FIGS. **15**A to **20**D illustrate a method for manufacturing an integrated circuit device **100** at various stages in accordance with some embodiments of the present disclosure. It is understood that additional operations may be provided before, during, and after the operations shown by FIGS. **15**A to **20**D, and some of the operations described below can be replaced or eliminated for additional embodiments of the method. The order of the operations/processes may be interchangeable.

(46) Reference is made to FIGS. **15**A-**15**C. FIG. **15**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **15**B is a cross-sectional view taken along line B-B of FIG. **15**A. FIG. **15**C is a cross-sectional view taken along line C-C of FIG. **15**A. STI features **130** are formed interposing the fins **112**, and the fins **112** extend above the STI features **130**. Dummy gate structure DG are formed over portions of the fins **112**. For better illustration, the semiconductor fins **112** are respectively labelled as semiconductor fins **112***a*-**112***d* in the present embodiments. In some embodiments, the semiconductor fins **112***a*-**112***d* are equidistantly arranged. For example, a distance between the fins **112***a* and **112***b* is substantially equal to a distance between the semiconductor fins **112***b* and **112***c* and a distance between the semiconductor fins **112***c* and **112***d*. Other details for forming the structure of FIGS. **15**A-**15**C are similar to those aforementioned in FIGS. **6**A-**9**D, and therefore not repeated herein.

(47) Reference is made to FIG. **16**. FIG. **16** is a cross-sectional view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure, and FIG. **16** is taken along the same cut as in FIG. **15**B. A spacer material layer **160** is conformally deposited over the structure of FIGS. **15**A-**15**C. For example, the spacer material layer **160** extends over top surfaces of the STI features **130**, sidewalls of the fins **112***a*-**112***d*, and top surfaces of the fins **112***a*-**112***d*. The spacer material layer **160** may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN films, silicon oxycarbide, SiOCN films, and/or combinations thereof. By way of example, the spacer material layer **160** may be formed by depositing a dielectric material using processes such as, CVD process, a subatmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a PVD process, or other suitable process.

(48) In some embodiments, a patterned mask P**2**′ is formed over a portion of the spacer material layer **160** over the fins **112***b* and **112***c*, and another portion of the spacer material layer **160** over the fins **112***a* and **112***d* may be free of coverage of the patterned mask P**2**′. The patterned mask P**2**′ may be a photoresist for protecting the spacer material layer **160** against subsequent etching process. The patterned mask P**2**′ may be formed by photolithography patterning processes, including photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), and/or other applicable processes. In some other embodiments, the patterned mask P**2**′ may be a hard mask for protecting the spacer material layer **160** against subsequent etching process.

(49) Reference is made to FIGS. **17**A-**17**C. FIG. **17**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some

embodiments of the present disclosure. FIG. **17**B is a cross-sectional view taken along line B-B of FIG. **17**A. FIG. **17**C is a cross-sectional view taken along line C-C of FIG. **17**A. The spacer material layer **160** (referring to FIG. **16**) is patterned to form spacers **161-162** on opposite sidewalls of the fin **112***b* and spacers **163-164** on opposite sidewalls of the fin **112***c*. The patterning process may include suitable etching process. By the etching process, the portions of the spacer material layer **160** free of coverage of the patterned mask P**2**′ (referring to FIG. **16**) is removed and etched away, while a portion of the spacer material layer **160** covered by the patterned mask P**2**′ (referring to FIG. **16**) is protected from being etched. The remaining portion of the spacer material layer **160** (referring to FIG. **16**) may be referred to as a spacer, which includes the spacers **161-164**.
(50) In the present embodiments, the remaining portion of the spacer material layer **160** (referring to FIG. **16**) further forms a portion **169** extending horizontally over the ST features **130** and connecting between the spacers **162** and **163**. In some other embodiments, the patterned mask P**2**′ (referring to FIG. **16**) may be designed in another way such that the portion **169** is etched away, and the spacer **162** may be spaced apart from the spacer **163**. After the patterning process, the patterned mask P**2**′ (referring to FIG. **16**) may be removed by suitable stripping process.
(51) In the present embodiments, the sidewalls of the fins **112***a* and **112***d* facing the semiconductor fins **112***b* and **112***c* may be free of a fin sidewall spacer. In alternative embodiments, other fin sidewall spacers may be formed on sidewalls of the fins **112***a* and **112***d* facing the semiconductor fins **112***b* and **112***c* with a top lower than that of the spacers **161-164**.
(52) Reference is made to FIGS. **18**A-**18**C. FIG. **18**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **18**B is a cross-sectional view taken along line B-B of FIG. **18**A. FIG. **18**C is a cross-sectional view taken along line C-C of FIG. **18**A. Exposed portions of the semiconductor fins **112** are etched by using the dummy gate structure DG, the gate spacers **150**, and the STI features **130** as an etch mask, resulting in recesses **112***r* into the semiconductor fins **112**. In some embodiments, the etching process may be a dry etching, a wet etch, or the combination thereof. In some embodiments, the portion **169** (referring to FIG. **17**B) may be removed by the etching process, such that the spacer **162** and **163** are disconnected from each other. In some embodiments, the etching process may also lower top surfaces of the spacer **161-164**.
(53) Reference is made to FIGS. **19**A-**19**C. FIG. **19**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **19**B is a cross-sectional view taken along line B-B of FIG. **19**A. FIG. **19**C is a cross-sectional view taken along line C-C of FIG. **19**A. Epitaxial features **171-174** are respectively formed over the exposed portions of the semiconductor fins **112***a*-**112***d*. In some embodiments, the epitaxial features **171-174** may include Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, or other suitable material. The epitaxial features **171-174** may be in-situ doped during the epitaxial process by introducing doping species including: p-type dopants, such as boron or BF.sub.2; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. If the epitaxial features **171-174** are not in-situ doped, an implantation process (i.e., a junction implant process) is performed to dope the epitaxial features **171-174**.
(54) The epitaxial features **171-174** may be formed by performing an epitaxial growth process that provides an epitaxial material on the fins **112***a*-**112***d*. Suitable epitaxial processes include CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy, and/or other suitable processes. The epitaxial growth process may use gaseous and/or liquid precursors, which interact with the composition of semiconductor materials of the fins **112***a*-**112***d*.
(55) In some further embodiments, the epitaxial growth of the epitaxial feature **172** is confined by the fin sidewall spacers **161** and **162**, and the epitaxial growth of the epitaxial feature **173** is confined by the fin sidewall spacers **163** and **164**. Through the confinement, the epitaxial features **172** and **173** are preventing from being merged with each other.

(56) In the present embodiments, the epitaxial growth of the epitaxial feature **171** is less or not confined by a fin sidewall spacer, such that the epitaxial feature **171** may extend laterally more than the epitaxial feature **173** extends. Through the configuration, the epitaxial feature **172** is merged with the epitaxial feature **171** and spaced apart from the epitaxial feature **173**.
(57) Similarly, in the present embodiments, the epitaxial growth of the epitaxial feature **174** is less or not confined by a fin sidewall spacer, such that the epitaxial feature **174** may extend laterally more than the epitaxial feature **172** extends. Through the configuration, the epitaxial feature **173** is merged with the epitaxial feature **174** and spaced apart from the epitaxial feature **172**.
(58) Reference is made to FIGS. **20**A-**20**D. FIG. **20**A is a schematic top view of the integrated circuit device **100** according to some embodiments of the present disclosure. FIG. **20**B is a cross-sectional view taken along line B-B of FIG. **20**A. FIG. **20**C is a cross-sectional view taken along line C-C of FIG. **20**A. FIG. **20**D is a cross-sectional view taken along line D-D of FIG. **20**A. An ILD layer **180** is formed on the substrate **110**. Subsequently, the dummy gate structure DG (referring to FIG. **19**A) is replaced with the metal gate structure **190**. Contacts **210** are formed in the ILD layer **180** and over the epitaxial features **171-174**. Other details of the present embodiments are similar to those mentioned in the embodiments of FIGS. **6**A to **14**D, and therefore not repeated herein.
(59) FIG. **21** is a cross-sectional view of an integrated circuit device **100** in accordance with some embodiments of the present disclosure, and FIG. **21** is taken along the same cut as in FIG. **20**B. The present embodiments are similar to the embodiments of FIGS. **15**A to **20**C, except that the patterning process performed to the spacer material layer **160** (referring to FIGS. **16-17**C) further form spacers **165-166** on opposite sidewalls of the fin **112***a*, and spacers **167-168** on opposite sidewalls of the fin **112***d*. In some embodiments, the height of the spacers **165-168** is less than the height of the spacers **162** and **163**. For example, top surfaces of the spacers **165-168** are at a position lower than top surfaces of the spacers **162-163**.
(60) Through the configuration, in the present embodiments, the epitaxial growth of the epitaxial feature **171** is less confined by the fin sidewall spacers **165-166** than the epitaxial growth of the epitaxial feature **173** being confined by the fin sidewall spacers **163-164**. Therefore, the epitaxial feature **171** may extend laterally more than the epitaxial feature **173** extends, which in turn result in that the epitaxial feature **172** is merged with the epitaxial feature **171** and spaced apart from the epitaxial feature **173**. Similarly, in the present embodiments, the epitaxial growth of the epitaxial feature **174** is less confined by the fin sidewall spacers **167-168** than the epitaxial growth of the epitaxial feature **172** being confined by the fin sidewall spacers **161-162**. Therefore, the epitaxial feature **174** may extend laterally more than the epitaxial feature **172** extends, which in turn result in that the epitaxial feature **173** is merged with the epitaxial feature **174** and spaced apart from the epitaxial feature **172**.
(61) In some embodiments, the height of the spacers **161** and **164** is less than the height of the spacers **162** and **163**, for example, being substantially equal to the height of the spacers **165-168**. For example, top surfaces of the spacers **161** and **164** are at a position lower than top surfaces of the spacers **162-163**, and substantially at the same level as top surfaces of the spacers **165-168**. The lower spacer **161** allows the epitaxial feature **172** to extend laterally toward the epitaxial feature **171**, which is beneficial for the merge between the epitaxial features **171** and **172**. Similarly, the lower spacer **164** allows the epitaxial feature **173** to extend laterally toward the epitaxial feature **174**, which is beneficial for the merge between the epitaxial features **173** and **174**. In some alternative embodiments, the height of the spacers **161** and **164** is greater than the height of the spacers **165-168**, for example, being substantially equal to the height of the spacers **162** and **163**. For example, top surfaces of the spacers **161** and **164** are at a position higher than top surfaces of the spacers **165-168**, and may be substantially at the same level as top surfaces of the spacers **162-163**. Other details of the present embodiments is similar to those mentioned above, and therefore not repeated herein.

(62) FIGS. **22** to **27**D illustrate a method for manufacturing an integrated circuit device **100** at various stages in accordance with some embodiments of the present disclosure. It is understood that additional operations may be provided before, during, and after the operations shown by FIGS. **22** to **27**D, and some of the operations described below can be replaced or eliminated for additional embodiments of the method. The order of the operations/processes may be interchangeable.

(63) Reference is made to FIG. **22**. FIG. **22** is a cross-sectional view taken of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure, and FIG. **22** is taken along the same cut as in FIGS. **10** and **16**. A spacer material layer **160** extends over top surfaces of the STI features **130**, sidewalls of the fins **112***a*-**112***d*, and top surfaces of the fins **112***a*-**112***d*. Other details regarding the structure of FIG. **22** and the method for forming the structure of FIG. **22** are similar to those mentioned above (e.g., in FIGS. **10** and **16**), and not repeated herein.

(64) In some embodiments, a patterned mask P**2**″ is formed over a portion of the spacer material layer **160**. The patterned mask P**2**″ may be a photoresist for protecting the spacer material layer **160** against subsequent etching process. The patterned mask P**2**′ may be formed by photolithography patterning processes, including photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), and/or other applicable processes. In some other embodiments, the patterned mask P**2**″ may be a hard mask for protecting the spacer material layer **160** against subsequent etching process.

(65) Reference is made to FIGS. **23**A-**23**C. FIG. **23**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **23**B is a cross-sectional view taken along line B-B of FIG. **23**A. FIG. **23**C is a cross-sectional view taken along line C-C of FIG. **23**A. The spacer material layer **160** (referring to FIG. **22**) is patterned to expose top surfaces of the fins **112***a*-**112***d*. The patterning process may include suitable etching process. By the etching process, portions of the spacer material layer **160** exposed by the patterned mask P**2**″ (referring to FIG. **22**) is etched away, while a portion of the spacer material layer **160** covered by the patterned mask P**2**″ (referring to FIG. **22**) is protected from being etched. The remaining portion of the spacer material layer **160** (referring to FIG. **22**) may be referred to as the spacer **160**′.

(66) After the patterning process, the spacer **160**′ may include spacers **161**-**168**, in which the spacers **161**-**162** are on opposite sidewalls of the fin **112***b*, the spacers **163**-**164** are on opposite sidewalls of the fin **112***c*, the spacers **165**-**166** are on opposite sidewalls of the fin **112***a*, and the spacers **167**-**168** are on opposite sidewalls of the fin **112***d*. In some embodiments, the spacer **160**′ further forms horizontal portions **169** extending horizontally over the ST features **130** and connecting between the spacers **162** and **163**, the spacers **166** and **161**, and the spacers **164** and **167**. In some other embodiments, the patterned mask P**2**″ (referring to FIG. **22**) may be designed in another way such that the horizontal portions **169** are etched away, which in turn will result in that the spacer **162** is spaced apart from the spacer **163**, the spacer **166** is spaced apart from the spacer **161**, and the spacer **164** is spaced apart from the spacer **167**. After the patterning process, the patterned mask P**2**″ (referring to FIG. **22**) may be removed by suitable stripping process. In some other embodiments, the patterning process may include suitable anisotropic etching process without using the patterned mask P**2**″ (referring to FIG. **22**) since the anisotropic etching process does not entirely remove the vertical portions the spacer material layer **160** (referring to FIG. **22**).

(67) Reference is made to FIGS. **24**A-**24**C. FIG. **24**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **24**B is a cross-sectional view taken along line B-B of FIG. **24**A. FIG. **24**C is a cross-sectional view taken along line C-C of FIG. **24**A. Exposed portions of the semiconductor fins **112***a*-**112***d* are etched by using the dummy gate structure DG, the gate spacers **150**, and the STI features **130** as an etch mask, resulting in recesses **112***r* into the

semiconductor fins **112***a*-**112***d*. In some embodiments, the etching process may be a dry etching, a wet etch, or the combination thereof. In some embodiments, the horizontal portions **169** (referring to FIG. **23**B) may be removed by the etching process, which in turn will result in that the spacer **162** is spaced apart from the spacer **163**, the spacer **166** is spaced apart from the spacer **161**, and the spacer **164** is spaced apart from the spacer **167**. In some embodiments, the etching process may also lower top surfaces of the spacer **161-168**.

(68) After the formation of the recesses **112***r*, a patterned mask P**3** may be formed over the semiconductor fins **112***b* and **112***c* and the spacers **161-164**, and expose the semiconductor fins **112***a* and **112***d* and the spacers **165-168**. The patterned mask P**3** may be a photoresist for protecting the semiconductor fins **112***b* and **112***c* against subsequent etching process. The patterned mask P**3** may be formed by photolithography patterning processes, including photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), and/or other applicable processes. In some other embodiments, the patterned mask P**3** may be a hard mask for protecting the spacers **161-164** against subsequent etching process.

(69) Reference is made to FIGS. **25**A-**25**C. FIG. **25**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **25**B is a cross-sectional view taken along line B-B of FIG. **25**A. FIG. **25**B is a cross-sectional view taken along line B-B of FIG. **25**C. The exposed portions of the semiconductor fins **112***a* and **112***d* are further recessed by suitable etching process, thereby deepening the recesses **112***r* (referring to FIG. **24**B) in the **112***a* and **112***d*. The deepened recesses **112***r* (referring to FIG. **24**B) in the fins **112***a* and **112***d* are referred to as recesses **112**C hereinafter. The etching process may use the dummy gate structure DG, the gate spacers **150**, the STI features **130**, and the patterned mask P**3** as an etch mask. In some embodiments, the etching process may be a dry etching, a wet etch, or the combination thereof. In some embodiments, the spacers **165-168** may be etched and/or removed by the etching process. Through the recessing portions of the semiconductor fins **112***a* and **112***d*, a top surface of the recessed portions of the semiconductor fins **112***a* and **112***d* is lower than a top surface of recessed portions of the second semiconductor fins **112***b* and **112***c*.

(70) In some other embodiments, the patterned mask P**3** (referring to FIG. **24**B) may be formed over the semiconductor fins **112***b* and **112***c* and the spacers **162** and **163**, and expose the semiconductor fins **112***a* and **112***d* and the spacers **161** and **164-168**. In some other embodiments, the spacers **161** and **164-168** may be etched and/or removed by the etching process.

(71) Reference is made to FIGS. **26**A-**26**C. FIG. **26**A is a schematic top view of the integrated circuit device at an intermediate stage of the manufacturing method according to some embodiments of the present disclosure. FIG. **26**B is a cross-sectional view taken along line B-B of FIG. **26**A. FIG. **26**C is a cross-sectional view taken along line C-C of FIG. **26**A. Epitaxial features **171-174** are respectively formed over the exposed portions of the semiconductor fins **112***a*-**112***d*. In some embodiments, the epitaxial features **171-174** may include Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, or other suitable material. The epitaxial features **171-174** may be in-situ doped during the epitaxial process by introducing doping species including: p-type dopants, such as boron or BF.sub.2; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. If the epitaxial features **171-174** are not in-situ doped, an implantation process (i.e., a junction implant process) is performed to dope the epitaxial features **171-174**.

(72) The epitaxial features **171-174** may be formed by performing an epitaxial growth process that provides an epitaxial material on the fins **112***a*-**112***d*. Suitable epitaxial processes include CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy, and/or other suitable processes. The epitaxial growth process may use gaseous and/or liquid precursors, which interact with the composition of semiconductor materials of the fins **112***a*-**112***d*.

(73) In the present embodiments, since the recesses **112***r*′ is deeper than the recess **112***r*, sizes of the epitaxial features **171** and **174** grown from the recess **112***r*′ are greater than sizes of the epitaxial features **172** and **173** grown from the recess **112***r*. For example, widths and heights of the epitaxial features **171** and **174** are greater than widths and heights of the epitaxial features **172** and **173**, resepectively. In some other embodiments, top surfaces of the epitaxial features **171** and **174** may be higher than top surfaces of the epitaxial features **172** and **173**. Through the configuration, the epitaxial feature **172** is merged with the epitaxial feature **171** and spaced apart from the epitaxial feature **173**, and the epitaxial feature **173** is merged with the epitaxial feature **174** and spaced apart from the epitaxial feature **172**.
(74) In some embodiments, the epitaxial growth of the epitaxial feature **172** is confined by the fin sidewall spacers **161** and **162**, and the epitaxial growth of the epitaxial feature **173** is confined by the fin sidewall spacers **163** and **164**. Through the confinement, the epitaxial features **172** and **173** are preventing from being merged with each other. In some embodiments, the epitaxial growth of the epitaxial feature **171** is less or not confined by a fin sidewall spacer, such that the epitaxial feature **171** may extend laterally more than the epitaxial feature **173** extends. Through the configuration, the epitaxial feature **172** is merged with the epitaxial feature **171** and spaced apart from the epitaxial feature **173**. Similarly, in the present embodiments, the epitaxial growth of the epitaxial feature **174** is less or not confined by a fin sidewall spacer, such that the epitaxial feature **174** may extend laterally more than the epitaxial feature **172** extends. Through the configuration, the epitaxial feature **173** is merged with the epitaxial feature **174** and spaced apart from the epitaxial feature **172**.
(75) Reference is made to FIGS. **27**A-**27**D. FIG. **27**A is a schematic top view of the integrated circuit device **100** according to some embodiments of the present disclosure. FIG. **27**B is a cross-sectional view taken along line B-B of FIG. **27**A. FIG. **27**C is a cross-sectional view taken along line C-C of FIG. **27**A. FIG. **27**D is a cross-sectional view taken along line D-D of FIG. **27**A. An ILD layer **180** is formed on the substrate **110**. Subsequently, the dummy gate structure DG is replaced with the metal gate structure **190**. Contacts **210** are formed in the ILD layer **180** and over the epitaxial features **171-174**. Other details of the present embodiments are similar to those mentioned above, and therefore not repeated herein.
(76) FIG. **28** is a cross-sectional view of an integrated circuit device in accordance with some embodiments of the present disclosure, and FIG. **28** is taken along the same cut as in FIG. **27**B. The present embodiments are similar to the embodiments of FIGS. **22** to **27**D, except that the etching process performed for deepening the recess **112***r* (referring to FIGS. **24**A-**25**C) does not fully consume the spacers **165** and **166** on opposite sidewalls of the fin **112***a* and spacers **167** and **168** on opposite sidewalls of the fin **112***d*. For example, in the present embodiments, the spacers **165-168** remains on the sidewalls of the fins **112***a* and **112***d*. In some embodiments, the etching process performed for deepening the recess **112***r* (referring to FIGS. **24**A-**25**C) may partially consume the spacers **165-168**, thereby lowering tops of the spacers **165-168**. For example, the tops of the spacers **165-168** are at a position lower than tops of the spacers **162** and **163**. In some embodiments, the height of the spacers **165-168** is less than the height of the spacers **162** and **163**.
(77) In the present embodiments, the epitaxial growth of the epitaxial feature **171** is less confined by the fin sidewall spacers **165-166** than the epitaxial growth of the epitaxial feature **173** being confined by the fin sidewall spacers **163-164**. Therefore, the epitaxial feature **171** may extend laterally more than the epitaxial feature **173** extends. Through the configuration, the epitaxial feature **172** is merged with the epitaxial feature **171** and spaced apart from the epitaxial feature **173**.
(78) Similarly, in the present embodiments, the epitaxial growth of the epitaxial feature **174** is less confined by the fin sidewall spacers **167-168** than the epitaxial growth of the epitaxial feature **172** being confined by the fin sidewall spacers **161-162**. Therefore, the epitaxial feature **174** may extend laterally more than the epitaxial feature **172** extends. Through the configuration, the epitaxial

feature **173** is merged with the epitaxial feature **174** and spaced apart from the epitaxial feature **172**. Other details of the present embodiments are similar to those mentioned above, and therefore not repeated herein.
(79) Based on the above discussions, it can be seen that the pre sent disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. One advantage is that through the configuration of fin sidewall spacers, the profile of the epitaxial features can be adjusted free from pitch limit, such that an epitaxial feature can be prevented from merging with an adjacent epitaxial feature. Another advantage is that the profile of an epitaxial feature can be asymmetric for merging with an adjacent epitaxial feature and spaced apart from another adjacent epitaxial feature. Still another advantage is that depths of recesses in fins would influence the size of the epitaxial features, such that a epitaxial feature cane be merged with an adjacent epitaxial feature and spaced apart from another adjacent epitaxial feature.
(80) According to some embodiments, an integrated circuit device includes a semiconductor substrate, first and second semiconductor fins over the semiconductor substrate, and first and second epitaxy structures respectively on the first and second semiconductor fins. The first epitaxy structure is merged with the second epitaxy structure, and a bottom surface of the second epitaxy structure is lower than a bottom surface of the first epitaxy structure.
(81) According to some embodiments, an integrated circuit device includes a semiconductor substrate; first to third semiconductor fins over the semiconductor substrate, wherein the second semiconductor fin is between the first and third semiconductor fins; first to third epitaxy structures respectively on the first to third semiconductor fins, wherein the second epitaxy structure is spaced apart from the first epitaxy structure, and the second epitaxy structure is merged with the third epitaxy structure; a first fin sidewall spacer on a sidewall of the first epitaxy structure facing the second epitaxy structure; and a second fin sidewall spacer on a first sidewall of the second epitaxy structure facing the first epitaxy structure, wherein a sidewall of the third epitaxy structure facing the second epitaxy structure has a portion at a same level as the second fin sidewall spacer, and the portion of the sidewall of the third epitaxy structure is free of a fin sidewall spacer.
(82) According to some embodiments, an integrated circuit device includes a semiconductor substrate; first and second semiconductor fins over the semiconductor substrate; first and second epitaxy structures respectively on the first and second semiconductor fins, wherein the second epitaxy structure is spaced apart from the first epitaxy structure; and a first fin sidewall spacer on a first sidewall of the first epitaxy structure facing the second epitaxy structure, wherein a second sidewall of the first epitaxy structure opposite the first sidewall is free of a fin sidewall spacer.
(83) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit device, comprising: a semiconductor substrate; first and second semiconductor fins over the semiconductor substrate; and first and second epitaxy structures respectively on the first and second semiconductor fins, wherein the first epitaxy structure is merged with the second epitaxy structure, and a bottom surface of the second epitaxy structure is lower than a bottom surface of the first epitaxy structure.

2. The integrated circuit device of claim 1, further comprising: a first fin sidewall spacer on a sidewall of the first epitaxy structure facing the second epitaxy structure, wherein a sidewall of the second epitaxy structure facing the first epitaxy structure is free of a fin sidewall spacer.
3. The integrated circuit device of claim 1, further comprising: a first fin sidewall spacer on a sidewall of the first epitaxy structure facing the second epitaxy structure; and a second fin sidewall spacer on a sidewall of the second epitaxy structure facing the first epitaxy structure, wherein a top of the second fin sidewall spacer is lower than a top of the first fin sidewall spacer.
4. The integrated circuit device of claim 1, further comprising: a third semiconductor fin over the semiconductor substrate, wherein the first semiconductor fin is between the second and third semiconductor fins; and a third epitaxy structure on the third semiconductor fin, wherein the third epitaxy structure is spaced apart from the first and second epitaxy structures.
5. The integrated circuit device of claim 4, wherein the bottom surface of the second epitaxy structure is lower than a bottom surface of the third epitaxy structure.
6. The integrated circuit device of claim 4, further comprising: a first fin sidewall spacer on a sidewall of the first epitaxy structure facing the third epitaxy structure; and a third fin sidewall spacer on a sidewall of the third epitaxy structure facing the first epitaxy structure, wherein a sidewall of the second epitaxy structure facing the first epitaxy structure is free of a fin sidewall spacer.
7. The integrated circuit device of claim 4, further comprising: a first fin sidewall spacer on a sidewall of the first epitaxy structure facing the third epitaxy structure; a second fin sidewall spacer on a sidewall of the second epitaxy structure facing the first epitaxy structure; and a third fin sidewall spacer on a sidewall of the third epitaxy structure facing the first epitaxy structure, wherein a top of the second fin sidewall spacer is lower than a top of the first fin sidewall spacer and a top of the third fin sidewall spacer.
8. The integrated circuit device of claim 4, wherein a distance between the first and second semiconductor fins is substantially equal to a distance between the first and third semiconductor fins.
9. An integrated circuit device, comprising: a semiconductor substrate; first to third semiconductor fins over the semiconductor substrate, wherein the second semiconductor fin is between the first and third semiconductor fins; first to third epitaxy structures respectively on the first to third semiconductor fins, wherein the second epitaxy structure is spaced apart from the first epitaxy structure, and the second epitaxy structure is merged with the third epitaxy structure; a first fin sidewall spacer on a sidewall of the first epitaxy structure facing the second epitaxy structure; and a second fin sidewall spacer on a first sidewall of the second epitaxy structure facing the first epitaxy structure, wherein a sidewall of the third epitaxy structure facing the second epitaxy structure has a portion at a same level as the second fin sidewall spacer, and the portion of the sidewall of the third epitaxy structure is free of a fin sidewall spacer.
10. The integrated circuit device of claim 9, wherein the first sidewall of the second epitaxy structure is more vertical than the sidewall of the third epitaxy structure.
11. The integrated circuit device of claim 9, wherein a second sidewall of the second epitaxy structure facing the third epitaxy structure has a portion at a same level as the second fin sidewall spacer, and the portion of the second sidewall of the second epitaxy structure is free of a fin sidewall spacer.
12. The integrated circuit device of claim 9, further comprising: a third fin sidewall spacer on a sidewall of the third epitaxy structure, wherein the third epitaxy structure extends beyond the third fin sidewall spacer.
13. The integrated circuit device of claim 9, wherein the second epitaxy structure does not extend beyond the second fin sidewall spacer.
14. The integrated circuit device of claim 9, wherein the first epitaxy structure does not extend beyond the first fin sidewall spacer.

15. An integrated circuit device, comprising: a semiconductor substrate; first and second semiconductor fins over the semiconductor substrate; first and second epitaxy structures respectively on the first and second semiconductor fins, wherein the second epitaxy structure is spaced apart from the first epitaxy structure; and a first fin sidewall spacer on a first sidewall of the first epitaxy structure facing the second epitaxy structure, wherein a second sidewall of the first epitaxy structure opposite the first sidewall of the first epitaxy structure is free of a fin sidewall spacer.

16. The integrated circuit device of claim 15, further comprising: a second fin sidewall spacer on a first sidewall of the second epitaxy structure facing the first epitaxy structure, wherein a second sidewall of the second epitaxy structure opposite the first sidewall of the second epitaxy structure is free of a fin sidewall spacer.

17. The integrated circuit device of claim 15, wherein the first sidewall of the first epitaxy structure is more vertical than the second sidewall of the first epitaxy structure.

18. The integrated circuit device of claim 15, wherein the first epitaxy structure does not extend beyond the first fin sidewall spacer.

19. The integrated circuit device of claim 15, wherein the second sidewall of the first epitaxy structure is an angled sidewall.

20. The integrated circuit device of claim 15, further comprising: a first contact over the first epitaxy structure; and a second contact over the second epitaxy structure.